

**A
Project Report
On
" Customer Preference and Trend Analytics for Retail Sector"**

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Under the guidance of
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Submitted at



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Chandubhai S. Patel Institute of Technology

At: Changa, Dist: Anand – 388421

October 2025

DECLARATION BY THE CANDIDATE

I hereby declare that the project report entitled “**Customer Preference and Trend Analytics for Retail Sector**” submitted by me to Chandubhai S. Patel Institute of Technology, Changa in partial fulfilment of the requirements for the award of the degree of **B.Tech Computer Science and Engineering**, from Department of Computer Science and Engineering, CSPIT, FTE, is a record of bonafide CSE401 Minor Project carried out by me under the guidance of **Prof. Akshita Kadam**. I further declare that the work carried out and documented in this project report has not been submitted anywhere else either in part or in full and it is the original work, for the award of any other degree or diploma in this institute or any other institute or university.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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To,
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Sub : Internship of Mr. Pranav Patel.

Dear Sir,

This is to confirm that **Mr. Pranav Mukeshkumar Patel** (Aadhar No. xxxx xxxx 4227) is undergoing an Internship with our organization since July 7, 2025 an will continue till October 31, 2025.. He has been assigned projects on Data Analytics and Business Intelligence. The undersigned, Mr. Divyang Shah (divyang@electrosoftglobal.com) is mentoring and supervising the projects assigned to him.

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Yours faithfully,
For **ELECTROSOFT**

Authorised Signatory

Place : Vadodara.
Date : September 27, 2025.

CERTIFICATE

This is to certify that the report entitled “**Customer Preference and Trend Analytics for Retail Sector**” is a bonafied work carried out by **PRANAV MUKESHKUMAR PATEL (22CS060)** under the guidance and supervision of **Prof. Akshita Kadam & Mr. Divyang Shah** for the subject **Project Minor (CSE401)** of 7th Semester of Bachelor of Technology in **Computer Science and Engineering** at Faculty of Technology & Engineering – CHARUSAT, Gujarat.

To the best of my knowledge and belief, this work embodies the work of candidate himself, has duly been completed, and fulfills the requirement of the ordinance relating to the B.Tech. Degree of the University and is up to the standard in respect of content, presentation and language for being referred to the examiner.

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ABSTRACT

This "Customer Preference and Trend Analytics for Retail Sector" project responds to the industry's pressing challenge of effectively utilizing large, disparate transactional data sets for strategic business decisions. The overall methodology entailed an end-to-end analytical process run on more than 293,000 retail transactions. The first step emphasized thorough cleaning and preprocessing of the data in Python. Later, debugging SQL queries were employed to expose and measure key data integrity problems, namely duplicate customer demographic attributes assigned to single Customer IDs. This called for the use of a solid Star Schema data model in Power BI through the use of Power Query to create clean Dimension Tables that corrected data anomalies and provided accurate analytics. The analysis was then conducted over four interrelated dashboards, culminating in a holistic Root Cause Analysis (RCA). Chief findings are the finding of a particular, high-spending Repeat Customer segment (18-25 year-old males with dual product affinities), the conclusion that a 7.3% sales decline in February results from an actual reduction in customer traffic, and the identification of a vital operational deficiency whereby 14% of negative complaints are due to quality problems after delivery as opposed to speed of fulfillment. The overall goal of this project was successfully achieved by shaping defective source information into strategic, actionable suggestions for maximizing customer loyalty, stabilizing revenue, and managing operational risk.

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"We may not achieve everything we dream, but we cannot achieve anything unless we dream." –

Pranav Patel
(22CS060)

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CHAPTER 1: INTRODUCTION

1.1 PROJECT SUMMARY

This project, Customer Preference and Trend Analytics for Retail Sector, represents an end-to-end data analysis and business intelligence solution developed from a raw transactional dataset of over 293,000 retail records. The primary goal was to transcend descriptive statistics and provide a comprehensive analytical system capable of generating actionable insights to influence business strategy. The workflow spanned four critical phases: data cleaning and feature engineering in Python, diagnostic querying using SQL, resolution of core data integrity issues through robust Star Schema modeling in Power BI, and the development of four interconnected dashboards. The final deliverables include strategic recommendations focused on optimizing customer loyalty, managing seasonal sales dips, and addressing operational flaws identified through root cause analysis (RCA).

1.2 PURPOSE

The purpose of this project is to create a reliable and repeatable analytical framework for the retail sector. The solution addresses the fundamental industry problem where large volumes of transaction data are often inconsistent and not utilized for strategic decision-making. The project serves to validate the importance of modern Data Modeling techniques (Star Schema) in resolving real-world data quality issues and to showcase the conversion of raw data into impactful business intelligence (BI) solutions. This framework ensures that data analysis becomes proactive rather than reactive, enabling management to anticipate trends like seasonal sales drops. By enforcing data integrity through structural modeling, the project minimizes the risk of making flawed business decisions based on inaccurate reporting. Ultimately, the purpose is to equip stakeholders with a visual, dynamic tool to monitor performance and gain immediate clarity on customer behavior, product success, and service quality.

1.3 OBJECTIVE

The system aims to achieve the following core analytical and structural objectives:

- **Data Integrity Resolution:** To identify and resolve critical inconsistencies within the raw data (e.g., conflicting customer demographics) by engineering a structural fix (Star Schema model) in Power BI.

- **Loyalty Profile Identification:** To quantify the repeat purchase rate and precisely define the demographic and behavioral profile of the most valuable loyal customers.
- **Revenue Stability Analysis:** To perform Root Cause Analysis (RCA) on the 7.3% February sales dip and determine if the cause is customer traffic or low spending.
- **Operational Risk Mitigation:** To investigate the source of high customer dissatisfaction and determine if the 14% 'Bad' feedback is due to fulfillment speed or to post-delivery product quality issues.
- **BI Development:** To develop four interconnected, interactive Power BI dashboards and strategic KPIs using DAX for continuous, dynamic performance monitoring.
- **Product Strategy Optimization:** To compare product performance across Volume vs. Value segments, determining the strategic role of top-selling high-volume brands (e.g., Pepsi) versus top-revenue high-value categories (e.g., Electronics).
- **Analytical Accuracy Validation:** To ensure the final integrity of the solution by implementing and validating core business metrics (KPIs) using the DAX language and cross-checking numerical results with diagnostic SQL queries.
- **Time-of-Day Behavior Analysis:** To identify operational flaws and customer engagement patterns by analyzing transaction volume and feedback concentration across engineered time slots (Morning, Afternoon, Evening, Night).
- **Data Preparation Mastery:** To demonstrate superior data preparation by cleaning, transforming, and exporting the final validated dataset (df_time_cleaned) using Python (Pandas) ready for subsequent analytical modeling.

1.4 SCOPE

The scope of this project is strictly defined by the end-to-end analytical workflow for the retail transactional dataset. It covers data cleaning and feature engineering using Python (Pandas) to create the validated analytical dataset. The scope includes SQL diagnostic querying and the full Star Schema modeling process in Power BI to ensure data integrity and structural accuracy. Furthermore, it covers detailed Root Cause Analysis (RCA) and the generation of strategic business recommendations based on the analytical findings. The final deliverable is the creation of four interactive Power BI Dashboards for strategic

management visualization. The project does not cover real-time data streaming, predictive modeling (ML/AI), or the physical implementation of the business recommendations.

1.5 TECHNOLOGY REVIEW

Analytical and Development Technologies:

- Python (Pandas, Matplotlib, Seaborn): Utilized extensively for Phase 1 (Data Cleaning) and Phase 2 (EDA). Pandas provided high-performance data structures necessary for efficient handling of the 293,000+ transaction records, including feature engineering (creating Time_Slot). Matplotlib and Seaborn were the primary tools for generating all statistical visualizations for the exploratory analysis.
- SQL (Structured Query Language): Used as a diagnostic and validation tool during Phase 3. Simple and complex SQL queries were essential for performing pre-modeling checks, verifying data types, and running aggregate tests to cross-validate final metrics calculated in Power BI, thus ensuring the numerical accuracy of the findings.
- Microsoft Power BI (Power Query and DAX): The core platform for Phase 3 (Data Modeling) and Phase 4 (Reporting). Power Query was used to perform the necessary transformations to build the Star Schema and resolve data inconsistencies. DAX (Data Analysis Expressions) was the language employed to author dynamic KPIs and strategic measures (e.g., Repeat Purchase Rate).

Data Modeling and Literature Review:

- Dimensional Modeling (Star Schema Architecture): The project's structural design adheres to the principles of dimensional modeling advocated by Ralph Kimball. This architecture mandates separating quantitative measures (stored in the Fact table) from descriptive text (attributes). This approach was critical for resolving the observed data integrity issue (conflicting customer demographics), ensuring that all calculated segments were based on a single, validated record for each customer, thereby preventing data misrepresentation.
- Analytical Framework (CRISP-DM): The project execution followed an adaptation of the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, moving logically from Business Understanding, Data Understanding,

Data Preparation, Evaluation, and finally, Deployment (Reporting). This disciplined approach ensured comprehensive project coverage and traceability.

CHAPTER 2: PROJECT MANAGEMENT

2.1 PROJECT PLANNING

2.1.1 Project Development Approach and Justification

The project utilized an Iterative and Incremental Analytical Approach, adhering closely to the principles of the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology. This adaptive framework was fundamentally necessary because the full scope of data quality issues within the transactional records was not known at the project's inception, making a fixed-scope approach unfeasible.

The process began with the initial data cleaning and Exploratory Data Analysis (EDA). This discovery stage critically revealed a severe data inconsistency issue—where conflicting demographic attributes were tied to single customer IDs. This unexpected finding necessitated an immediate and unplanned pivot to the structural development stage, requiring the design and meticulous implementation of the Star Schema in Power BI to structurally resolve the data flaws.

This methodology was chosen because standard analysis could not proceed without first fixing the unreliable data foundation. The development cycle followed a clear pattern: Analysis, Evaluation and Model Refinement. This disciplined process ensured that all final analytical dashboards and strategic metrics were built upon a foundation of validated and accurate data, guaranteeing the overall reliability of the solution for management decisions.

2.1.2 Project Effort and Time, Cost Estimation

Project Duration: 12 weeks (July 2025 - September 2025)

Table 2. 1 Project Duration

Week	Phase	Key Activities
Week 1-2	Phase 1: Setup & Data Prep	Project initialization, raw data review, Python scripting setup, initial cleaning plan design.
Week 3	Phase 1: Cleaning Deep Dive	Data Cleaning Core: Handled missing values, removed duplicates, standardized date/time formats, and imputed key fields (Age, Ratings)

Week	Phase	Key Activities
Week 4	Phase 1: Feature Engineering	Finalized data export of df_time_cleaned. Engineered new features (Hour, Time_Slot) required for time-based analysis.
Week 5-6	Phase 2: EDA & Discovery	Conducted statistical summaries and Exploratory Data Analysis (EDA). Generated core visualizations (e.g., Monthly Sales Trends, Top Categories).
Week 7-8	Phase 2: RCA Diagnostics	Completed all 14+ visualizations (Heatmap, Donut Chart). Identified and documented the Data Inconsistency issue and the February sales dip.
Week 9	Phase 3: SQL & Validation	Executed diagnostic SQL queries to quantify data inconsistencies. Validated analytical metrics against raw data aggregates.
Week 10-11	Phase 3: Modeling & DAX	Designed and implemented the Star Schema model in Power BI. Created clean Dimension Tables (Customer, Product) using Power Query. Authored dynamic DAX measures (KPIs).
Week 12	Phase 4: Reporting & Finalization	Developed four interactive Power BI Dashboards. Completed Root Cause Analysis (RCA). Finalized project documentation and PPT presentation

Effort Estimation:

- Total working hours: Approximately 350 hours
- Average daily effort: 3-4 hours (alongside college schedule)
- Peak effort during module development: 5-6 hours daily (during Phase 3 Modeling and Phase 4 Dashboarding)
- Reduced effort during exam periods: 1-2 hours daily

Project Completion Status:

- Month 1 (July): 20% completed (Phase 1 Finalized)
- Month 2 (August): 45% completed (Phase 2 Finalized)
- Month 3-4 (Sept/Oct): 100% completed (Phase 3 & 4 Finalized)

Cost Estimation:

The project utilized entirely free resources, resulting in zero monetary cost:

- Analytical tools (Python, Pandas, Matplotlib): Free (Open-Source)
- BI Software (Microsoft Power BI Desktop): Free

- Database Querying (SQL): Free (Open-Source)

2.1.3 Roles and Responsibilities

This project was developed by a single team member who handled all aspects of development. The various roles and responsibilities managed throughout the project are listed below.

Requirements Analyst

- Identified core business objectives for the retail dataset.
- Analyzed stakeholder needs (e.g., identifying the cause of the sales dip, profiling loyal customers).
- Defined the functional requirements for the four Power BI dashboards.

Data Engineer

- Handled the data preparation and cleaning pipeline using Python (Pandas).
- Managed missing values, data type conversions, and duplicate removal.
- Engineered new analytical features, such as the Time_Slot column.
- Exported the final validated df_time_cleaned dataset.

Data Architect

- Designed the Star Schema data model to resolve data integrity issues.
- Created the clean Dimension Tables (Customer, Product, Calendar) in Power Query.
- Defined and managed the relationships between the Fact Table and Dimension Tables in Power BI.

BI Developer and Data Analyst

- Designed the user interface and user experience (UI/UX) for the four Power BI dashboards.
- Authored all key performance indicators (KPIs) and metrics using the DAX language.
- Conducted the Exploratory Data Analysis (EDA) and the final Root Cause Analysis (RCA).
- Translated numerical findings into actionable business recommendations.
- Integrated frontend with backend APIs

Quality Analyst

- Executed diagnostic SQL queries to validate the integrity of the source data.

- Cross-checked final DAX measures against SQL aggregates to ensure numerical accuracy.
- Tested dashboard functionality, including slicers, filters, and cross-report interactions.
- Validated that the final model correctly fixed the data inconsistency errors.

Documentation Writer

- Prepared the monthly project progress reports for July, August, and October.
- Wrote this final comprehensive project report.
- Created the final PowerPoint (PPT) presentation deck for the viva defense.

2.1.4 Group Dependencies

This project was developed independently by a single student, eliminating internal team coordination needs. However, several critical technical and procedural dependencies influenced the development process and analytical workflow.

Data Pipeline Dependencies

- The raw retail dataset first had to be rigorously cleaned using Python (Phase 1) to create the `df_time_cleaned` file.
- The Exploratory Data Analysis (EDA) in Phase 2 was fully dependent on this cleaned dataset to generate initial insights.
- The discovery of the data inconsistency issue during EDA was a critical dependency that triggered the need for advanced modeling.
- Changes to core tables required updating dependent tables and APIs

Data Modelling Dependencies

- The entire analytical solution (Phase 4 Dashboards) was critically dependent on the successful design and implementation of the Star Schema in Phase 3.
- The creation of clean Dimension Tables (Customer, Product) had to be completed before any accurate DAX measures could be written.
- A failure in the data model or its relationships would have rendered all final visualizations and KPIs incorrect and invalid.

Validation Dependencies

- The use of diagnostic SQL queries was a procedural dependency required to confirm and quantify the data integrity flaws before committing to the Star Schema solution.

- Final DAX measures (like Total Revenue) were validated against SQL aggregate queries to ensure numerical accuracy.

External Tool Dependencies

- The project was dependent on Python and its libraries (Pandas, Matplotlib) for all data preparation and initial exploration.
- The project relied entirely on Microsoft Power BI Desktop for the data modeling (Power Query) and dashboard development (DAX) phases.
- The entire analysis was dependent on the availability and structural soundness of the initial raw dataset.

2.2 PROJECT SCHEDULING

Week 1-2 (July 2025):

- Kicked off the project by gathering all the files and figuring out the main goals.
- Did an initial review of the huge 290,000-record dataset to see what we were working with.
- Set up the Python environment (using Pandas and NumPy) and designed the initial plan for data cleaning.

Week 3-4 (July 2025):

- This was the main data cleaning phase. I focused on handling all the missing values for key columns like Age and Ratings.
- Removed all the duplicate transactions to make sure our numbers wouldn't be inflated.
- Standardized tricky columns like Date and engineered the new Time_Slot feature for our analysis.
- Finished this phase by exporting the final df_time_cleaned file.

Week 5-6 (August 2025):

- Started Phase 2 (EDA). I began by running statistical summaries to understand the data's shape.
- Created the first set of major visualizations, like the Monthly Sales Trend line chart and the Top 10 Product Categories bar chart.

Week 7-8 (August 2025):

- Finished all 14+ visualizations for the EDA, including the customer loyalty donut chart and the feedback-rating heatmap.

- During this deep dive, I made the most critical discovery: the data inconsistency issue where customer IDs had conflicting information.
- User experience improvements based on mentor feedback

Week 9-10 (September 2025):

- Began Phase 3 (Modeling). I used SQL queries to double-check and prove the data inconsistency I found in the EDA.
- Started designing the Star Schema data model in Power BI as the solution to this problem.
- Began building the clean 'Dimension Tables' (like Customer_Table and Product_Table) using Power Query.

Week 11-12 (September 2025):

- Finalized the complete Star Schema model in Power BI, connecting all the tables.
- Wrote all the key DAX measures (like KPIs for Total Revenue and Repeat Purchase Rate).
- Built all four interactive Power BI dashboards and performed the final Root Cause Analysis.
- Prepared the final project report (this document) and the viva presentation (PPT).

CHAPTER 3: SYSTEM REQUIREMENTS STUDY

3.1 USER CHARACTERISTICS

The final Power BI report is designed for business managers and analysts inside the retail company. Most users just need basic computer skills—they should know how to use a mouse, click on filters, and read charts to understand the insights.

- **Executive (Senior Management):** This user wants the big picture, fast. They will mainly look at the Global Sales Overview dashboard to see total revenue, monthly sales trends, and overall performance. They need clear, high-level KPIs to make major strategic decisions.
- **Marketing Analyst:** This user needs to dig deep into customer behavior. They will use the Customer Insights dashboard to understand the 68% repeat rate and identify the exact profile of the loyal customer. They will also use the Product Performance dashboard to see the "Volume vs. Value" (Pepsi vs. Electronics) dynamic and plan new promotions.
- **Operations Manager:** This user is focused on service quality. They will spend their time on the Operational & Feedback dashboard. Their main goal is to understand the root cause of the 14% 'Bad' feedback and see why issues are happening on 'Delivered' orders, so they can fix the process.
- **Data Analyst (Technical User):** This is the role I performed. This user needs to be able to check the data model, validate the Star Schema, and ensure the DAX calculations are accurate. Their job is to make sure the report is reliable and trustworthy for everyone else.

3.2 HARDWARE AND SOFTWARE REQUIREMENTS

3.2.1 Development Requirements

Hardware Requirements:

- **Processor:** Intel Core i5 or better (or equivalent modern CPU) for efficient data processing.

- **RAM:** Minimum 8 GB is essential; 16 GB is highly recommended for smoother operation when handling the large 290,000+ record dataset in Python and Power BI concurrently.
- **Hard Disk:** At least 10 GB of free space required for installing software, storing the large raw data files, and exporting multiple cleaned versions.
- **Monitor:** Standard resolution of at least 1024 x 768 to comfortably manage the multi-pane interfaces of Power BI and the Python environment.

Software Requirements:

- **Operating System:** Windows 10/11 or macOS, as these fully support the required BI and analytical tools.
- **Analytical Environment:** Python 3.8+ runtime environment.
- **Key Python Libraries:** Pandas (for data manipulation), NumPy (for numerical operations), and Matplotlib/Seaborn (for EDA visualizations).
- **Business Intelligence Tool:** Microsoft Power BI Desktop (for data modeling, DAX scripting, and dashboard development).
- **Database Tool:** A basic SQL Client (like SQLite or a command-line tool) is needed for running diagnostic validation queries.
- **Code Editor:** Visual Studio Code or any code editor for managing Python scripts.

3.2.2 User Requirements

- **Viewing Application:** Access via a modern web browser (Chrome, Firefox, or Edge) if the report is published to the Power BI Service, or the Power BI Mobile App for remote access.
- **Software Compatibility:** JavaScript and Cookies enabled in the browser for secure authentication and full interactivity of the published report.
- **Connection:** A stable internet connection is required to reliably access the cloud-hosted Power BI report.
- **Report Access:** Users must have the necessary Power BI user license (if applicable) and permissions to view the shared report workspace.

3.3 ASSUMPTIONS AND DEPENDENCIES

Assumptions :

- **Data Integrity Baseline:** It is assumed that while demographic consistency was flawed, the core transactional metrics (e.g., Amount, Transaction_ID, Date) recorded in the raw data were fundamentally accurate.
- **Data Representativeness:** The analyzed retail dataset is assumed to be a fair representation of the company's customer behavior and sales patterns during the specified time period.
- **User Literacy:** End-users (Management and Analysts) are assumed to have basic analytical literacy required to interpret standard data visualizations and understand dashboard filters.
- **Software Availability:** It is assumed that the analytical team has access to the required local development tools, namely Python, Power BI Desktop, and an SQL Client.
- **Static Scope:** All analysis and recommendations are based solely on the fixed, historical dataset (df_time_cleaned) and are not expected to integrate real-time or future data.

Dependencies :

- **Modeling Dependency:** The entire project's analytical validity relies critically on the successful construction and integrity of the Star Schema data model built in Power BI (Phase 3).
- **Data Validation Dependency:** The accuracy of the findings is dependent on the ability to use SQL diagnostic queries to cross-validate key metrics and structural integrity before final report generation.
- **Tool Interoperability:** The sequential workflow requires accurate and successful transfer of data from Python (cleaning) , SQL (validation) , Power BI (modeling and visualization).
- **Customer Attribute Resolution:** The final segmentation and loyalty analysis are dependent on the logical and accurate resolution of the conflicting customer demographic attributes during the Power Query modeling process.

- **Mentor feedback** and college review are vital for continuous progress and improvements.

CHAPTER 4: SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEM

The 'Current System' for business intelligence in the retail environment often relies on static reporting and manual data extraction, leading to unreliable, reactive decision-making. Common analytical activities—such as tracking customer trends, profiling loyalty, and linking feedback to operational status—are hindered by flawed data and inefficient tools.

- **Data Quality Issues:** The core transactional data contains critical flaws, most notably conflicting customer demographic attributes tied to the same Customer IDs. This structural inconsistency makes reliable segmentation (by Age, Gender, Income) impossible without manual intervention.
- **Static Reporting:** Management typically receives slow, static reports (like basic Excel summaries) which lack the dynamic capability to filter, cross-analyze, and drill down into the root cause of a trend.
- **Hidden Analytical Gaps:** Key strategic insights, such as the true cause of the 7.3% February sales dip or the concentration of 'Bad' feedback on *Delivered* orders, are obscured or missed entirely by basic aggregation.
- **Manual Validation:** Any data checking or integrity verification requires laborious manual SQL scripting or cross-checking in spreadsheets, which is slow and unsustainable for a large dataset.
- **Reactive Decision-Making:** Managers must depend on end-of-period summaries. This makes decision-making slow and results unpredictable, as there is no easy way to analyze the true cost of operational failures or anticipate customer churn.

Because the data is used in this manual, flawed way, there is a higher chance of making strategic mistakes based on incorrect segments and unreliable metrics. The existing system severely lacks the data integrity and dynamic visualization required for proactive retail strategy.

4.2 PROBLEM AND WEAKNESSES OF CURRENT SYSTEM

- **Structural Data Flaws:** The most severe weakness is the pervasive data inconsistency issue within the core dataset, where single customers are associated

with multiple, conflicting demographic attributes (e.g., age, gender). This makes accurate customer segmentation and profiling unreliable.

- **Misleading Metrics and Reports:** When using simple spreadsheets, the data's structural flaws lead to miscalculated or misleading aggregate metrics. Managers risk making high-cost decisions (like launching a marketing campaign) based on inaccurate customer profiles or sales numbers.
- **Hidden Root Causes:** The system is unable to pinpoint the root cause of critical anomalies. For example, manual reporting can show a sales dip but cannot definitively tell management if it's due to fewer customers (traffic) or lower spending (AOV).
- **Untracked Operational Failures:** The link between customer feedback and specific operational events is missed. Critical issues, such as the 14% 'Bad' feedback being tied to 'Delivered' status (indicating a quality/packaging problem), remain hidden, preventing targeted operational fixes.
- **Reactive and Slow Reporting:** Generating complex reports (like the Repeat Purchase Rate broken down by Age Group) requires laborious, manual SQL or spreadsheet manipulation. This delays insights, forcing management to be reactive rather than proactive in their strategy.
- **Inability to Cross-Analyze:** The existing method lacks a centralized, Star Schema-based structure, making it impossible to dynamically link and analyze different data dimensions (e.g., linking a customer's income to their preferred payment method across regions) in real-time.

Ultimately, these core weaknesses severely reduce the reliability of insights and slow down the strategic response of the retail management team, increasing the chance of costly errors due to flawed analytical foundations.

4.3 REQUIREMENTS OF NEW SYSTEM

4.3.1 Functional Requirements

- **Data Integrity Enforcement:** Must ensure that the core data model resolves all conflicting customer demographic attributes through the Star Schema structure, providing a single, trustworthy source of truth for segmentation.

- **Dynamic Visualization:** Must allow end-users to dynamically cross-filter and drill down into all metrics across dimensions (e.g., viewing Product Performance filtered by Customer Age Group).
- **Root Cause Analysis Capability:** Must provide the necessary DAX measures and visuals to pinpoint the root cause of observed anomalies, such as determining if the February sales dip is due to traffic or spending.
- **Strategic Profile Generation:** Must clearly visualize and quantify the loyalty profile (demographics, purchasing habits) of the repeat customer segment.
- **Operational Insight Mapping:** Must display the direct link between 'Bad' customer feedback and Order Status to identify quality and packaging issues versus fulfillment delays.
- **Key Metric Generation:** Must accurately calculate and display core retail KPIs, including Repeat Purchase Rate, Average Order Value (AOV), and Revenue Share by product type and gender.
- **Time-of-Day Analysis:** Must incorporate the engineered Time_Slot feature to analyze customer behavior and error concentration based on time of day (Morning, Afternoon, Evening, Night).

4.3.2 Non-Functional Requirements

- **Reliability (Data Integrity):** The underlying Star Schema model must guarantee data integrity and accuracy, with zero tolerance for metric miscalculation or corruption.
- **Performance:** The Power BI dashboards must load and refresh quickly. Response time for filtering and cross-analysis should be under 3 seconds to support fast decision-making.
- **Usability:** The interface must be user-friendly and intuitive, requiring minimal training for management to operate the slicers and interpret the visualizations.
- **Accessibility:** The final report must be easily accessible, ideally published to the Power BI Service or viewable in a modern web browser without any local software installation.

- **Scalability:** The data model must be structured to easily accommodate future data growth (more transactions) and allow for the simple addition of new metrics or dimensions without complex reconstruction.
- **Security:** If published to a cloud service (Power BI Service), the report must support future implementation of Row-Level Security (RLS) to control which managers see which regions or product lines.

4.4 FEASIBILITY STUDY

4.4.1 Does the system contribute to the overall objectives of the organization?

Absolutely, the Business Intelligence (BI) system contributes directly to the retail organization's core objectives. It enhances efficiency by moving analysis away from slow, flawed static reports and toward dynamic, validated insights. Managers gain immediate clarity on critical areas like customer segmentation, operational failures, and revenue trends, enabling faster and better data-driven decisions. By clearly identifying the loyal customer profile and the root cause of sales dips, the system supports targeted marketing and cost-saving efforts, directly contributing to the company's growth and profitability.

4.4.2 Can the system be implemented using the current technology and within the given cost and schedule constraints?

Yes, the system is highly feasible. It utilizes a modern, yet widely supported and entirely free analytical stack: Python for data preparation, SQL for validation, and Microsoft Power BI Desktop for modeling and visualization. The entire project was planned and executed within the 14-week academic timeline, leveraging the student's expertise and the existing academic schedule. The tools chosen are mature, reliable, and require zero monetary licensing cost, ensuring smooth implementation well within any reasonable budget and time constraint.

4.4.3 Can the system be integrated with other systems which are already in place?

The BI solution is designed to be highly scalable and future-proof. While the 'current systems' are largely manual, the final output—a Star Schema data model—is the industry standard for analytical reporting. This standardized architecture allows for smooth integration in the future with broader enterprise systems, such as accounting software or

data warehouses. This design ensures that as the retail company evolves digitally, the core analytical model can connect to new data sources, providing flexibility and guaranteeing the long-term utility of the solution.

4.5 PROCESS IN NEW SYSTEM

The 'New System' is defined by the rigorous, multi-stage analytical process designed to transform raw transaction records into strategic business intelligence. The workflow below ensures data integrity and actionable outputs.

- **Data Ingestion and Cleaning (Python):** The process begins by loading the raw data into the Python environment. Scripts handle initial data cleaning, manage missing values, and engineer essential features like the Time_Slot. The output is the clean, validated df_time_cleaned file.
- **Diagnostic Validation (SQL):** Before modeling, SQL queries are run against the data to perform integrity checks. This phase is crucial for quantifying the severity of structural flaws, such as the conflicting customer demographic attributes.
- **Data Modeling (Power Query):** The cleaned data is loaded into Power BI. Power Query is used to structurally resolve the flaws by creating the Star Schema. This involves building clean Dimension Tables (Customer, Product) that serve as a single, consistent source of truth.
- **Metric Generation (DAX):** The modeled data is used to author complex DAX measures and key performance indicators (KPIs). These measures (e.g., Repeat Purchase Rate, AOV) provide the necessary quantitative metrics for the dashboards.
- **Insight Discovery (RCA):** The system enables analysts to dynamically cross-filter the four dashboards to perform Root Cause Analysis (RCA). This is where key findings are extracted, such as the cause of the February sales dip and the true source of 'Bad' feedback.
- **Visualization and Reporting:** The final output is the set of interactive Power BI Dashboards. These provide management with immediate, dynamic clarity on customer loyalty, product performance, and operational trends, enabling faster strategic decisions.

This standardized analytical process improves accuracy, validates the integrity of the core retail data, and ensures management receives reliable, proactive insights rather than slow, static reports.

4.6 FEATURE OF NEW SYSTEM

The new Business Intelligence (BI) system delivers several important analytical features that fundamentally improve retail management by streamlining the path from raw data to strategic decision-making:

- **Data Integrity Assurance:** The system's foundation is the Star Schema data model, which structurally resolves the conflicting customer demographics present in the raw data, ensuring all reported metrics are accurate and trustworthy.
- **Dynamic Loyalty Profiling:** Provides precise quantification of the Repeat Purchase Rate and offers the ability to dynamically segment customers by demographics, allowing the targeting of the high-value 18-25 age group.
- **Root Cause Dissection:** Features the analytical power to pinpoint the *true cause* of trends and anomalies, such as isolating the February sales dip to a customer traffic problem rather than lower spending.
- **Operational Risk Mapping:** Connects customer feedback to order status, clearly showing that the 14% 'Bad' feedback is due to post-delivery quality issues on *Delivered* orders, directing resources away from fulfillment speed.
- **Volume vs. Value Comparison:** Offers dedicated visualization comparing high-revenue categories (like Electronics) with high-volume brands (like Pepsi), enabling a strategic two-tier product marketing approach.
- **Advanced Analytical Metrics:** The solution includes meticulously authored DAX measures for complex KPIs like Average Order Value (AOV), Revenue Share by Gender, and detailed trend analysis across all four dashboards.
- **User-Friendly Interface:** The entire reporting suite is accessible via interactive Power BI dashboards, designed with a clean, unified interface that requires minimal training for managers to explore data dynamically.
- **Data Validation Transparency:** The analytical process includes built-in validation, where key metrics are consistently cross-checked against SQL

aggregates, guaranteeing the numerical fidelity of the data presented in the dashboards.

These features collectively transform slow, flawed reporting into a dynamic analytical environment, enabling faster and smarter management decisions based on validated insights.

4.7 FUNCTIONS OF NEW SYSTEM

The Business Intelligence solution is designed with specific functional outputs that empower managers to perform complex, validated analysis that was previously impossible.

- **Dynamic Data Filtering and Exploration:** The primary function is to allow users to dynamically cross-filter all four interconnected dashboards by any dimension (Time, Geography, Customer Segment, Product Category). This enables instantaneous, complex data slicing.
- **Root Cause Dissection:** The system facilitates Root Cause Analysis (RCA) by enabling metric isolation. Users can confirm the 7.3% February sales dip is strictly a customer traffic issue and not a spending problem.
- **Loyalty Profile Generation:** The BI solution functions to clearly quantify and visualize the specific demographic and purchasing profile of the loyal, repeat customer segment, allowing the marketing team to target lookalike audiences.
- **Operational Flaw Mapping:** A core function is to map customer dissatisfaction directly to the Order Status. This allows managers to see that the 14% 'Bad' feedback is concentrated on orders marked 'Delivered,' redirecting operational focus from slow shipping to product quality and packaging errors.
- **Analytical Metric Generation:** The dashboards function to display meticulously authored DAX KPIs, providing accurate and validated retail metrics like the Repeat Purchase Rate, Average Order Value (AOV), and Revenue Share by product type.
- **Data Integrity Visualization:** The system functions as a proof-of-concept for data integrity. All displayed metrics are visually guaranteed to be accurate because they are sourced from the structurally sound, Star Schema model, which resolved the flaws of the original source data.

4.8 CONTEXT DIAGRAM(DFD-0)

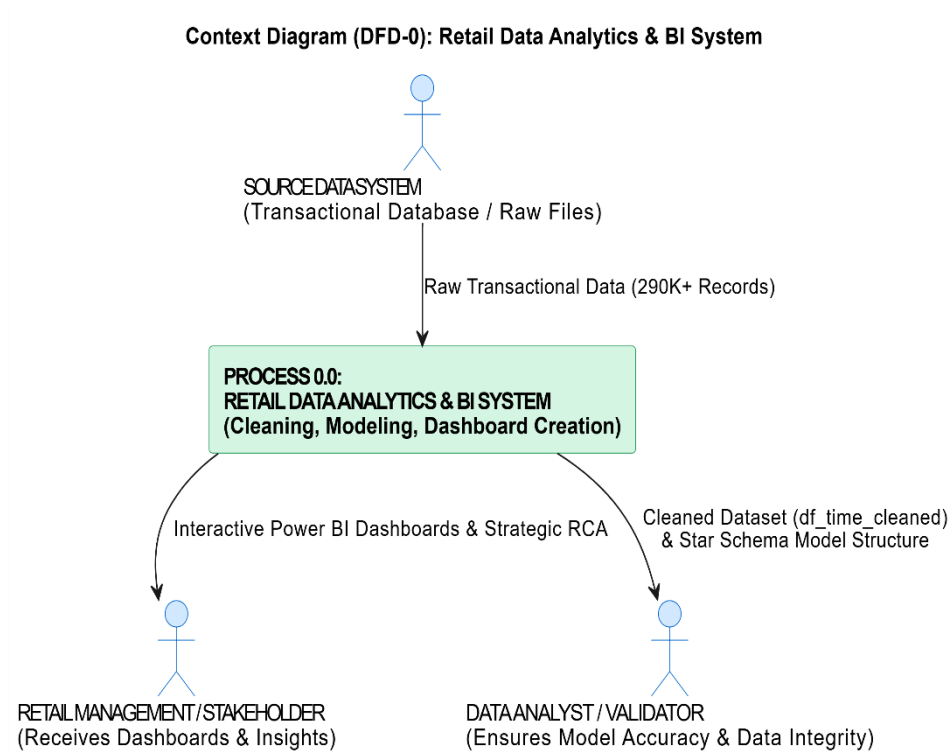


Fig 4. 1: Context Diagram

4.9 DFD – 1 and DFD - 2

DFD-1: Data Preparation & Validation (Retail Data Analytics & BI System)

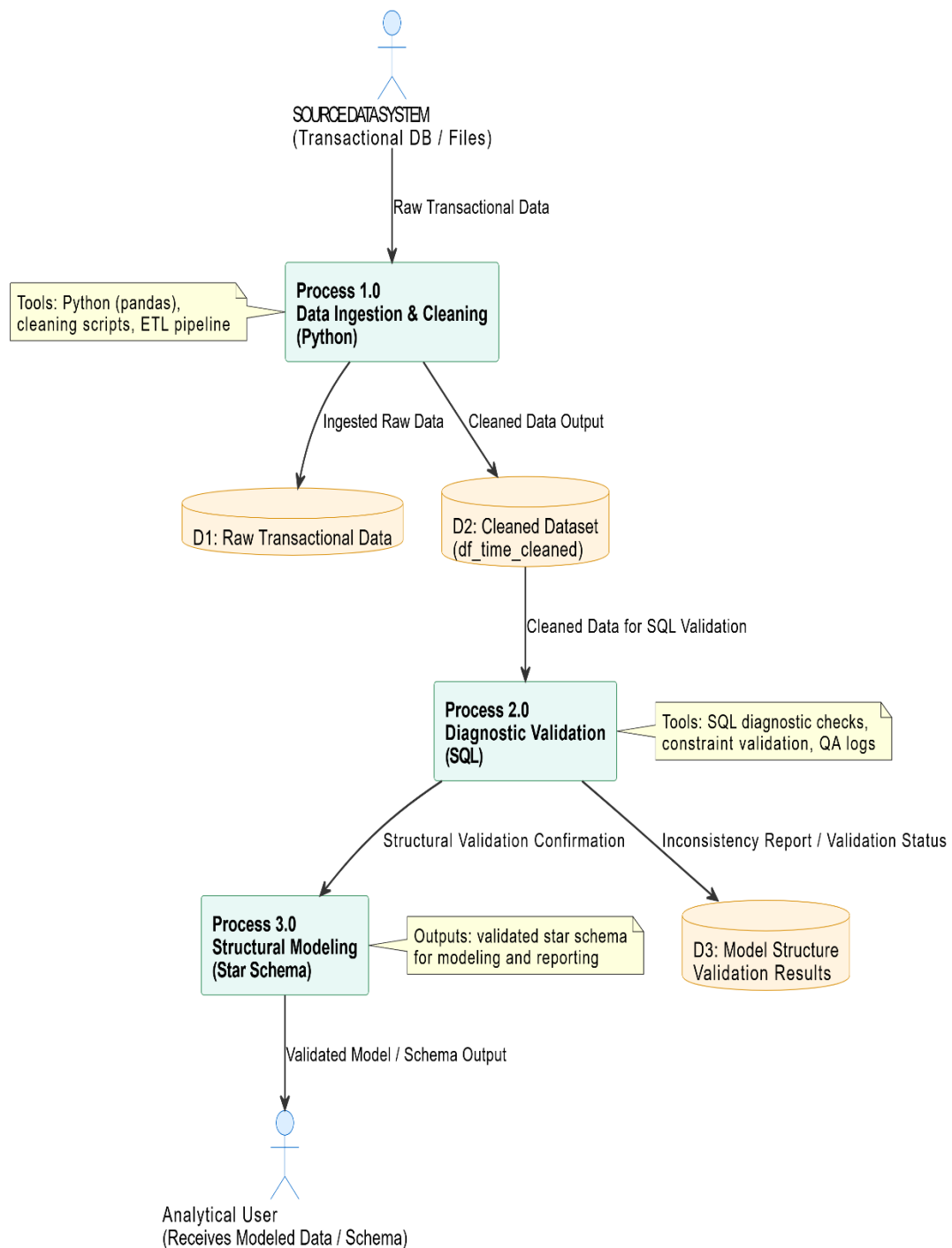


Fig 4. 2: DFD-1

DFD-2: Star Schema Modeling & Reporting (Process 3.0 Decomposed)

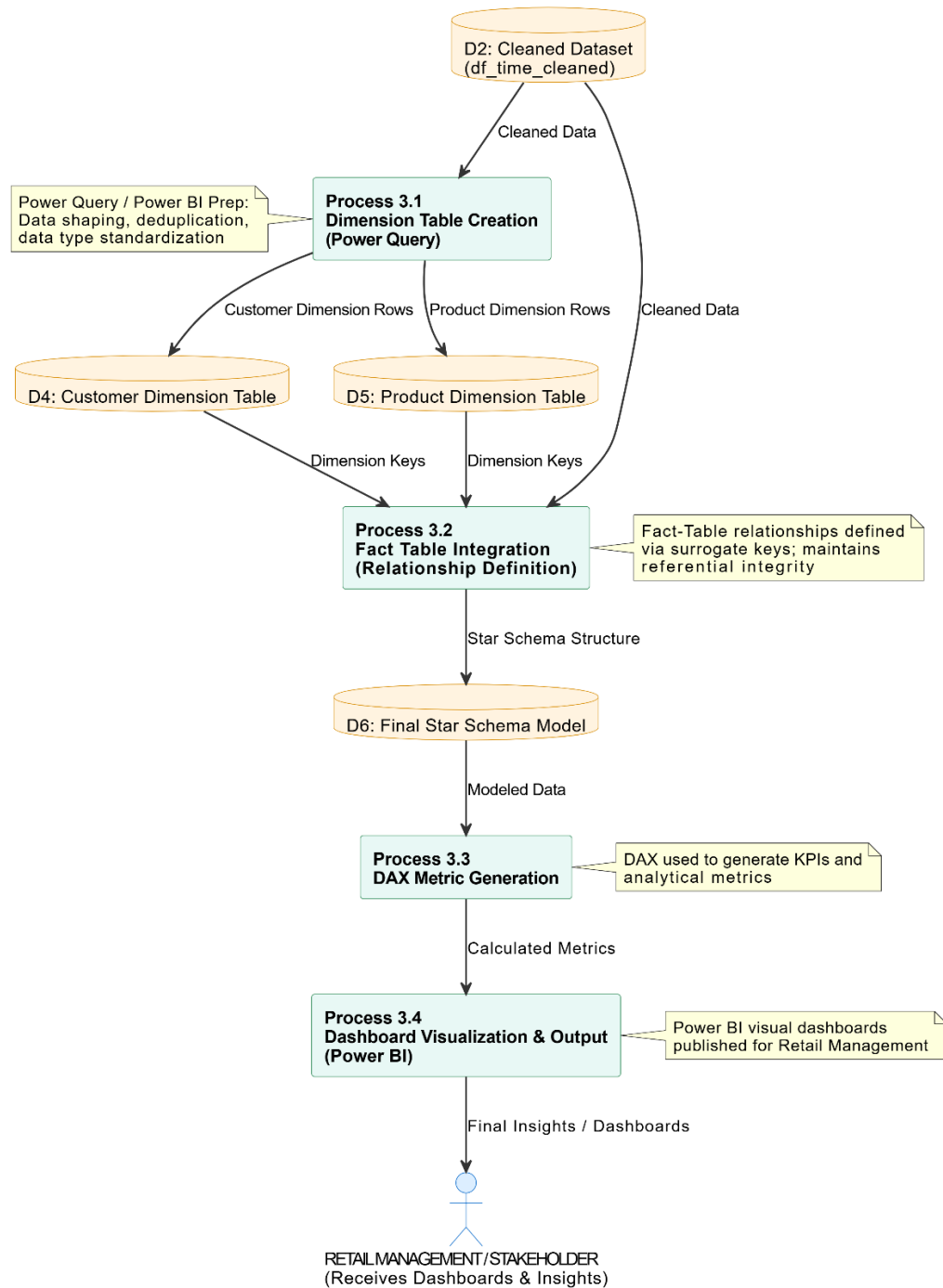


Fig 4. 3: DFD-2

4.10 DATA MODELLING

4.10.1 ER Diagram

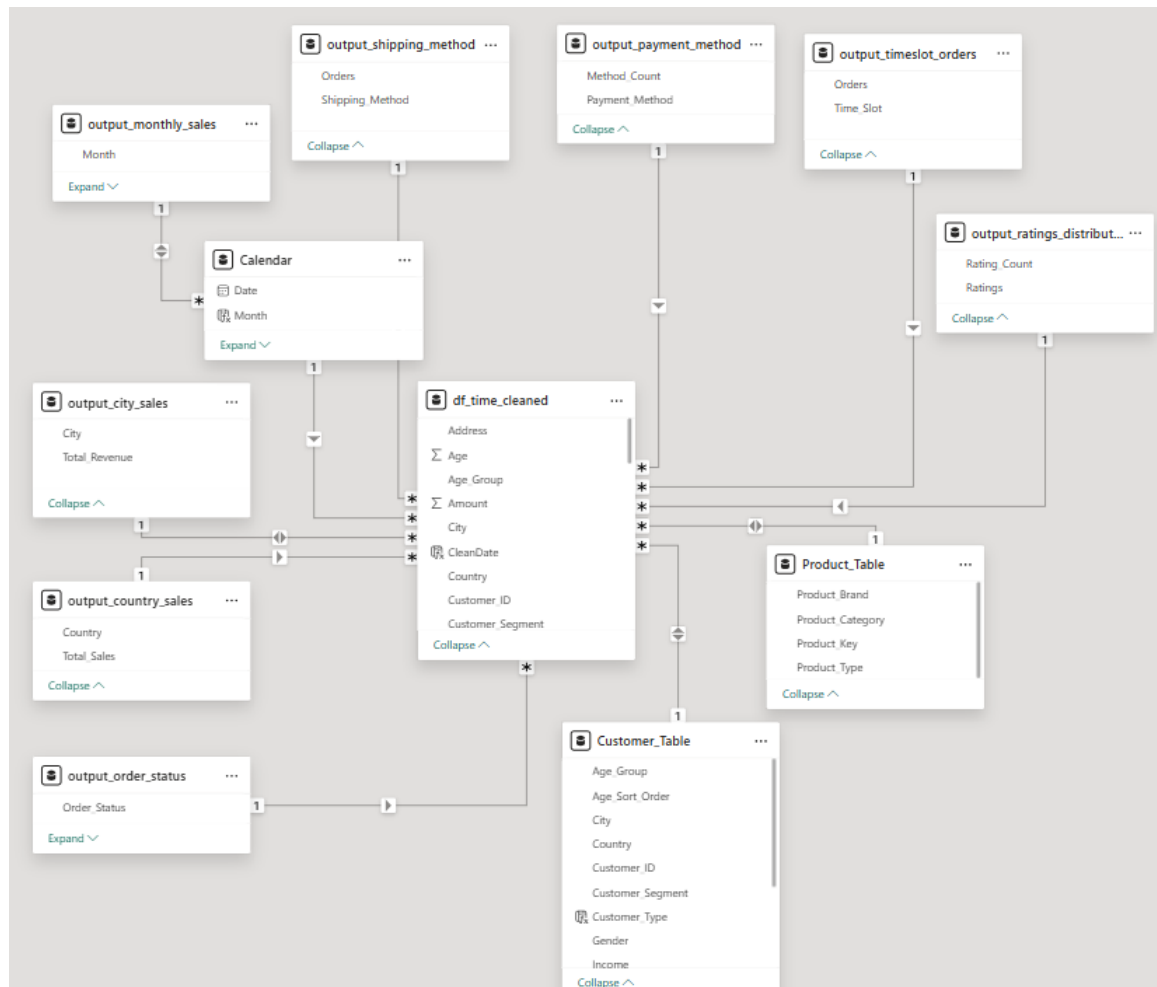


Fig 4. 4: ER Diagram

4.10.2 Data Dictionary

The Data Dictionary documents the metadata for the corrected analytical attributes, representing the single, verified source of truth established during the Power BI modeling process.

The Customer_Table (Dimension) defines the unique, corrected customer demographics. The primary key is Customer_ID (Integer), which is the unique identifier for each customer. The table includes Age_Group (Text), Gender (Text), and Income (Text). These three descriptive attributes are Corrected Attributes, meaning they were normalized using Power

Query grouping logic to resolve all conflicting values found in the raw data, ensuring they are consistent for every customer.

The Product_Table (Dimension) defines the product hierarchy. Its primary key is Product_Key (Text), which is a unique composite key identifying the specific product/brand combination. It includes Product_Brand (Text), Product_Category (Text)—the high-level category like Electronics or Grocery—and Product_Type (Text), which is the sub-classification of the product category.

The core transaction data resides in the df_time_cleaned (Fact Table). This table uses the unique transaction ID as its primary key. Key attributes include Amount (Decimal), Total_Purchases (Integer), and Foreign Keys linking to the dimensions: Customer_ID, Product_Key, and CleanDate (Date). This table contains the measures used for all financial aggregations in the report.

4.11 MODULES OF NEW SYSTEM

The Business Intelligence solution is structured into four functional modules. These modules ensure the data flows correctly from its raw state to the final strategic output, guaranteeing integrity and usability.

1. Data Preparation Module (Python)

This module covers all pre-analytical processing required to make the raw data fit for modeling. It primarily utilizes Python (Pandas) to handle the large transaction volume and structural cleaning. Key tasks include data type standardization, handling null values, removing duplicate records, and the engineering of analytical features, such as the crucial Time_Slot column for time-based analysis.

2. Data Modeling & Validation Module (SQL & Power Query)

This is the core integrity module. It utilizes SQL diagnostic queries for pre-modeling validation, followed by Power Query in Power BI to construct the Star Schema. The primary function of this module is to structurally resolve the conflicting customer demographic attributes by creating clean, singular records within the Dimension Tables (Customer, Product, Calendar).

3. Metrics & Analytics Module (DAX)

This module focuses on transforming the modeled data into quantifiable insights. It relies on the DAX (Data Analysis Expressions) language to author all complex business metrics and KPIs. Functions include calculating the Repeat Purchase Rate, determining Average Order Value (AOV), generating Revenue Share percentages, and setting up the analytical framework required for Root Cause Analysis (RCA).

4. Visualization & Reporting Module (Power BI Dashboards)

This module serves as the final output interface. It is composed of four interconnected Power BI Dashboards (Global Sales, Customer Insights, Product Performance, Operational & Feedback). Its function is to provide the end-user (management) with a dynamic, intuitive environment for exploration, filtering, and cross-analyzing data to inform strategic and operational decisions.

4.12 USE CASE DIAGRAM

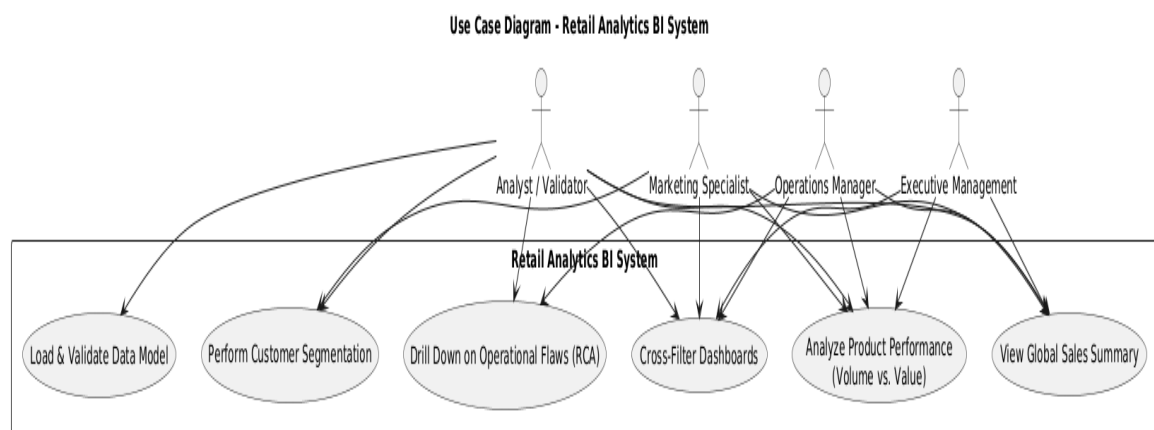


Fig 4. 5: Use Case Diagram

4.13 CLASS DIAGRAM

Class Diagram - Retail Analytics BI System (Star Schema Model)

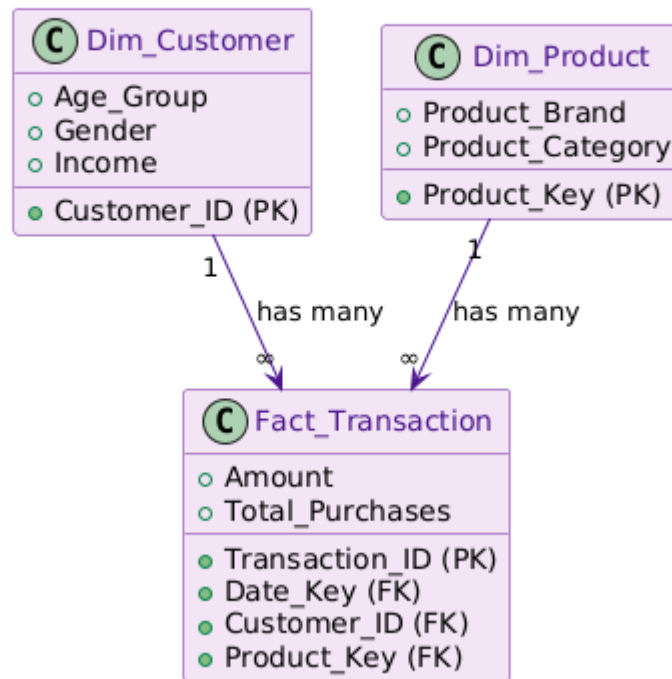


Fig 4. 6: Class Diagram

4.14 SEQUENCE DIAGRAM

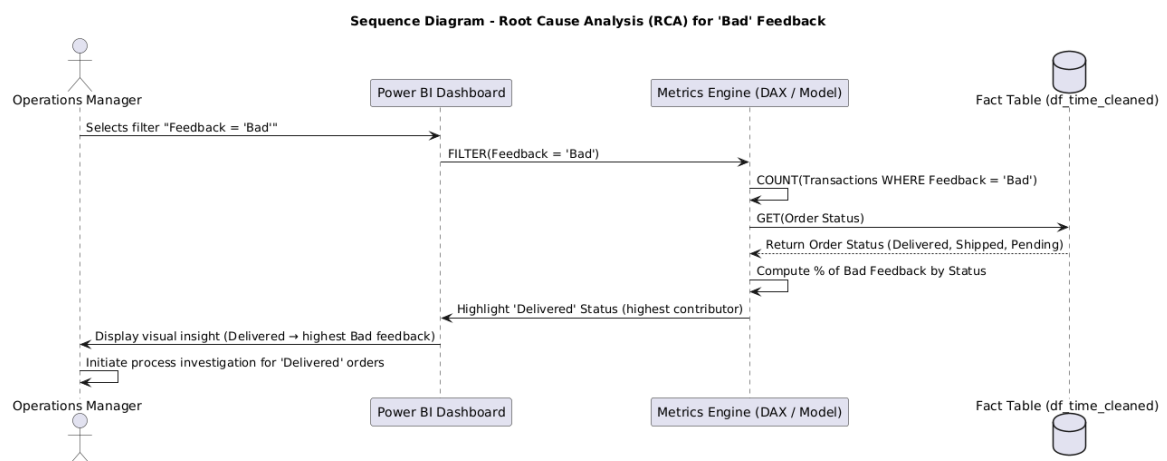


Fig 4. 7: Sequence Diagram

CHAPTER 5: SYSTEM DESIGN

5.1 SYSTEM APPLICATION DESIGN

The system application design is based on a structured, layered Business Intelligence (BI) Architecture rather than a traditional application stack. This design prioritizes data integrity, analytical performance, and visual communication.

The architecture is centered around the Star Schema data model, which sits at the core of the Power BI environment. This model acts as the single source of truth, having resolved all data inconsistencies found in the raw transactional files.

The front-end design provides four main analytical dashboards:

- **Global Sales Overview Dashboard:** This serves as the Executive view, providing high-level KPIs, monthly revenue trends, and geographic performance (used to analyze the February sales dip).
- **Customer Insights Dashboard:** This focuses on loyalty and segmentation, profiling the repeat customer base against demographics (Age, Income, Gender) and calculating the Repeat Purchase Rate.
- **Product Performance Dashboard:** This allows for strategic comparison, separating product performance into Volume (e.g., Pepsi brand) versus Value (e.g., Electronics category) dynamics.
- **Operational & Feedback Dashboard:** This is the service quality control center, mapping customer satisfaction against order status to identify that the 14% 'Bad' feedback is related to post-delivery quality issues.

The overall flow involves Python communicating with the data, the Power BI Engine executing the business logic (authored in DAX), and the Power BI Desktop Interface presenting the results to the user for interactive analysis. Data consistency and security checks are enforced at the Power Query modeling layer.

5.1.1 Method Pseudo Code

The project does not contain transactional backend code; therefore, this section details the pseudo-code for the DAX (Data Analysis Expressions) measure used to calculate the

Repeat Purchase Rate (RPR). This demonstrates complex analytical logic used within the Power BI Metrics Module.

5.1.2 Pseudo Code for Calculate Repeat Purchase Rate (RPR)

DAX Pseudo Code: [Repeat Purchase Rate %]

DEFINE MEASURE [Repeat Purchase Rate %] AS

```
// 1. Calculate the total count of unique customers from the clean dimension table.  
Total_Customers = COUNTROWS( Customer_Table )
```

```
// 2. Calculate the count of customers who have made 2 or more transactions.  
Repeat_Customers = CALCULATE( COUNTROWS( Customer_Table ), FILTER(  
VALUES( Customer_Table[Customer_ID] ), CALCULATE( COUNTROWS(  
df_time_cleaned ) ) >= 2 ) )
```

```
// 3. Return the final percentage. RETURN DIVIDE( Repeat_Customers,  
Total_Customers, 0 )
```

Key Logic and Analytical Purpose:

- The logic operates entirely on the structurally sound Star Schema model to ensure accuracy.
- It iterates over the unique customers in the clean Dimension Table (Customer_Table) to count individuals.
- It applies a powerful filter context to the Fact Table (df_time_cleaned) to classify a customer as 'Repeat' only if their total transaction count is 2 or more.
- The final result is the ratio of Repeat Customers to Total Customers, providing the core KPI for the Customer Insights Dashboard.

5.2 INPUT/OUTPUT AND INTERFACE DESIGN

5.2.1 Samples of Forms, Reports and Interface

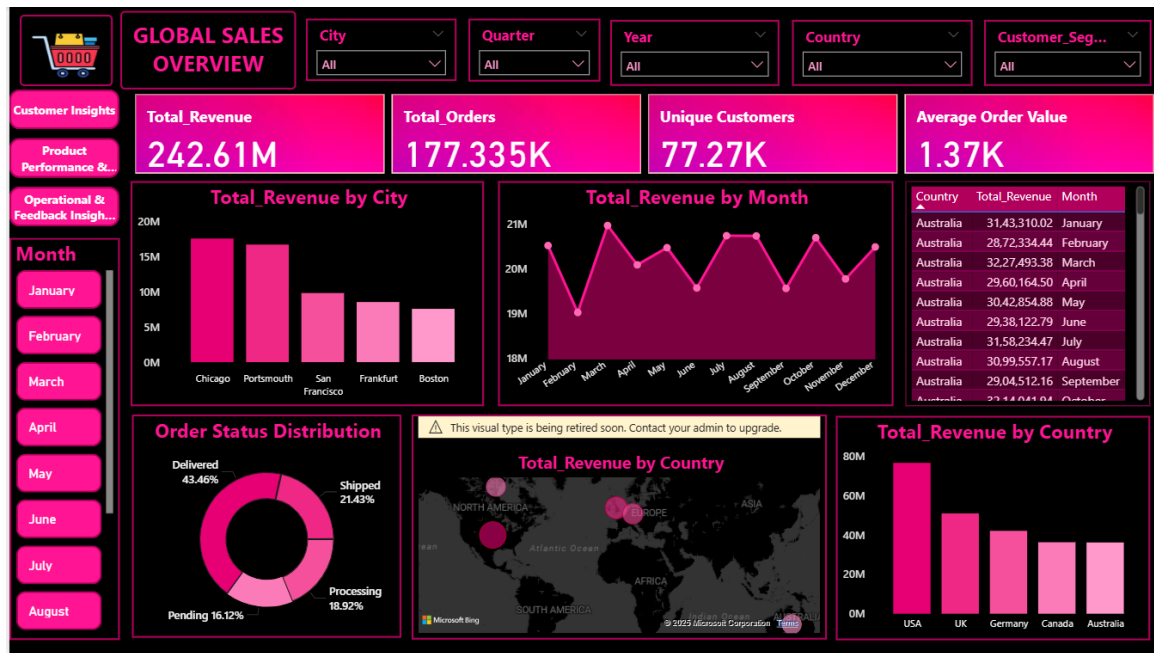


Fig 5. 1: Global Sales Overview Dashboard



Fig 5. 2: Customer Insights Dashboard

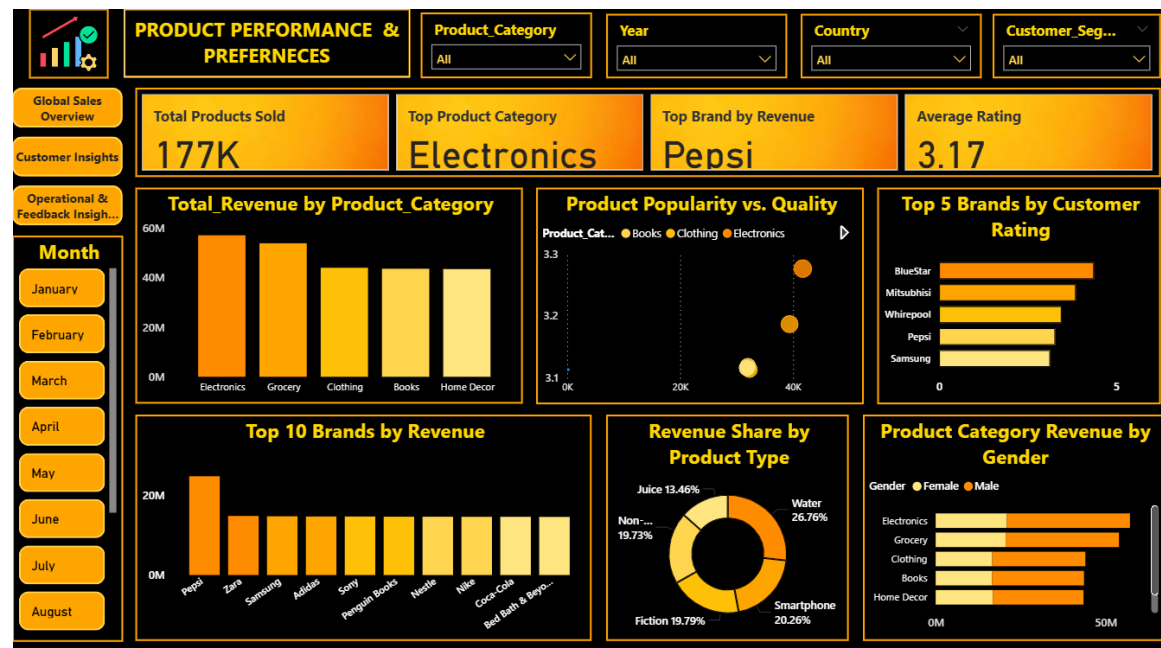


Fig 5. 3: Product Performance And Preferences Dashboard



Fig 5. 4: Operational And Feedback Insights Dashboard

5.2.2 Access Control and Security

Since this project focuses on Business Intelligence (BI), "security" centers on guaranteeing data integrity and controlling who sees sensitive information, rather than protecting against external attacks.

- **Data Integrity and Validation:** The most critical security feature of this system is the Star Schema model itself. By structurally resolving the conflicting customer demographics in Power Query, the model ensures that the data is protected from internal corruption and misreporting. This was cross-validated using SQL diagnostic queries to guarantee numerical accuracy.
- **Access Control (Row-Level Security - RLS):** The Power BI environment is designed to support future implementation of Row-Level Security (RLS). This advanced feature allows management to control data visibility, ensuring, for example, that a regional manager can only view the sales data specific to their geography.
- **Data Privacy:** Sensitive data attributes (like direct names or private emails, if available in the raw files) were anonymized or excluded during the initial Python cleaning phase (Phase 1), adhering to basic data privacy principles.
- **Version Control:** The analytical scripts (Python and DAX) are managed using Git, ensuring a complete history of changes and preventing unauthorized or untraceable modifications to the core analytical logic.

5.3 SYSTEM STRUCTURAL DESIGN

The System Structural Design for this project defines the layered architecture of the Business Intelligence (BI) solution. It outlines how data is transformed from its raw state into the final analytical model, replacing a traditional software application structure with a robust data pipeline.

- **Layer 1: Data Preparation (Python Layer):** This forms the base. Raw transactional data is ingested and processed using Python (Pandas). This layer is responsible for crucial tasks like data type standardization, imputation, and feature engineering (creating the Time_Slot), preparing the high-quality df_time_cleaned dataset.
- **Layer 2: Data Modeling (Power Query/Star Schema Layer):** This is the structural core of the system. Data is loaded into Power BI, where the flawed transactional data is structurally corrected by implementing the Star Schema. This layer creates the clean, centralized Fact Table and the denormalized Dimension Tables (Customer_Table, Product_Table), resolving the integrity issues found in the raw source.

- **Layer 3: Metrics Engine (DAX Layer):** This functional layer operates on the Star Schema. It utilizes the DAX language to define all complex business logic, calculated columns, and Key Performance Indicators (KPIs), such as the Repeat Purchase Rate and Average Order Value.
- **Layer 4: Visualization and Presentation (Dashboard Layer):** The final output layer consists of the four interactive Power BI Dashboards. This is the user-facing interface designed for dynamic exploration and strategic decision-making by retail management.

This layered structural design ensures that the system is modular, maintainable, and guarantees that any output presented in Layer 4 is built upon a validated, structurally sound foundation from Layers 1 and 2.

CHAPTER 6: IMPLEMENTATION PLANNING

6.1 IMPLEMENTATION ENVIRONMENT

The implementation environment for this project was chosen based on the efficiency and industry standards required for robust data analysis and Business Intelligence (BI) development. This architecture ensures reliability, structural integrity, and superior performance for handling large transactional data volumes.

- **Data Preparation Environment (Python):** The initial layer for cleaning and manipulation utilizes Python, specifically the Pandas library. Python's high-performance data processing capabilities were critical for handling the 290,000+ transaction records, ensuring efficient data transformation, cleaning, and feature engineering (like creating the Time_Slot column).
- **Modeling and Metrics Environment (Power BI):** The core analytical environment is Microsoft Power BI Desktop. This software provides the engine for structural design using Power Query (essential for building the Star Schema and resolving data inconsistencies) and the specialized language DAX (Data Analysis Expressions) for authoring complex business logic and analytical metrics (KPIs).
- **Validation Environment (SQL):** SQL (Structured Query Language) was used as a necessary diagnostic tool within the development pipeline. SQL provided the platform for executing quick aggregate queries and integrity checks, allowing the analyst to cross-validate metrics calculated in the Power BI model against the raw source data.
- **System Integrity:** The entire environment is focused on data integrity, ensuring that the outputs presented in the final report are reliable and validated. The Power BI layer supports dynamic filtering and interactive dashboards, which are critical for the efficient running of a strategic BI system.

6.2 MODULES SPECIFICATION

The Business Intelligence solution is logically divided into analytical and reporting modules to ensure data integrity, maintainability, and clear separation of functions within the analytical workflow. Each module contributes to the final strategic output.

1. Data Preparation Module

- Manages the initial data ingestion, cleaning, and preprocessing utilizing Python (Pandas).
- Handles missing values, standardizes date formats, removes duplicate transactions, and performs feature engineering (e.g., creating the Time_Slot column).
- Produces the high-quality, analysis-ready df_time_cleaned file.

2. Data Modeling & Integrity Module

- The core structural module that resolves data flaws by designing and implementing the Star Schema in Power BI.
- Utilizes Power Query to create clean Dimension Tables (Customer, Product, Calendar) that house unique, consistent attributes.
- Defines the essential relationships between the Fact Table and Dimension Tables, structurally guaranteeing data integrity.

3. Metrics Engine Module

- Translates complex business logic into quantifiable Key Performance Indicators (KPIs) using the DAX language.
- Calculates essential retail metrics such as Repeat Purchase Rate, Average Order Value (AOV), and strategic revenue share percentages.
- Provides the analytical foundation required for Root Cause Analysis (RCA).

4. Customer Insights Module (Dashboard)

- The first reporting module, designed to focus entirely on customer behavior and loyalty.
- Presents dynamic visuals and metrics for profiling the high-value, loyal customer segment against demographics (Age, Income, Gender).
- Serves as the primary interface for marketing analysts to plan retention campaigns.

5. Operational & Feedback Module (Dashboard)

- The quality control module designed for operations management.

- Maps customer satisfaction metrics (e.g., 14% 'Bad' feedback) directly against Order Status (Delivered, Shipped) to isolate service failures.
- Provides the necessary data for operations managers to address post-delivery quality issues and night-shift errors.

6. Product Performance Module (Dashboard)

- Focuses on the strategic role of products and brands.
- Compares the dual revenue stream: Volume (e.g., Pepsi brand) versus Value (e.g., Electronics category).
- Provides visuals for analyzing brand quality against popularity and identifying product strategy needs.

6.3 SECURITY FEATURES

Since this project delivers a business intelligence (BI) solution built upon a static data model, the concept of "security" focuses on ensuring data integrity, preventing unauthorized metric manipulation, and controlling report visibility, rather than traditional software security.

- **Data Integrity as Primary Security:** The most critical security feature of this system is the underlying Star Schema data model. This structure structurally resolves the fundamental flaw of conflicting customer demographics found in the raw data, thereby protecting the final analysis from miscalculation. This structural integrity is continuously verified using SQL diagnostic queries and DAX validation logic, ensuring metrics are numerically trustworthy.
- **Access Control and Visibility (RLS Support):** The Power BI environment is designed to support future implementation of Row-Level Security (RLS). This is a critical feature that allows management to define rules so that users (e.g., Regional Managers) can only view the data relevant to their specific clearance (e.g., their region's sales data), controlling access to sensitive financial information.
- **Data Privacy and Anonymization:** During the initial Python cleaning phase, any highly sensitive or direct personal identifiers (if available in the raw files) were anonymized or excluded, adhering to basic principles of data privacy.

- **Version Control for Logic:** All core analytical logic (Python cleaning scripts and DAX measures) are managed using Git. This ensures a complete audit trail, preventing untraceable or unauthorized modifications to the foundational analysis and metrics.

6.4 CODING STANDARD

To guarantee the quality, readability, and long-term maintainability of the project's analytical logic, strict standards were followed across all technological environments (Python, SQL, and DAX).

- **Consistent Naming Conventions:** PEP 8 conventions were adhered to for Python scripting, using descriptive snake_case for variables and functions. For the DAX language, clear naming conventions were used for all measures (e.g., [Total Revenue], [Repeat Purchase %]) to ensure the analytical structure is auditable.
- **Modular Structure:** The analytical workflow was organized into separate, sequential files for each major task (e.g., one Python file for cleaning, one Power BI file for modeling/DAX), ensuring that each step had a single responsibility and improving maintainability.
- **Commenting and Documentation:** Comprehensive inline comments were used within Python scripts to explain complex cleaning logic and within DAX measures to define calculation steps. This ensures easy traceability of the analytical pipeline.
- **Error Handling:** Analytical logic included robust error handling, such as using DIVIDE(numerator, denominator, 0) in DAX, to prevent crashes or misleading results when data inconsistencies (like division by zero) were encountered.
- **Validation Standards:** For SQL queries, a professional standard was maintained, using uppercase for all keywords (SELECT, FROM, WHERE) to improve readability and facilitate easy cross-checking of metrics against the modeled data.
- **Version Control Practices:** All primary analytical assets (Python files and the Power BI model file structure) were managed using Git. This practice ensures a complete, traceable history of all changes to the foundational data preparation and metric logic.

Adhering to these standards not only improved the quality of the analytical work but also laid a foundation for future scalability and easy auditing of the final BI solution.

CHAPTER 7: TESTING

7.1 TESTING PLAN

The testing plan for this project focused on guaranteeing the data integrity of the final analytical model and validating that all generated metrics are accurate and trustworthy. Testing was carried out across all major analytical stages (Cleaning, Modeling, and Visualization).

Types of Testing

- **Data Cleaning/Unit Testing:** Each function and script in the Python environment was tested independently to verify that it performed the required transformation as expected. This included ensuring the removal of duplicates and the successful standardization of Date and Time columns.
- **Data Integrity Testing:** This was the most critical testing type. It involved running SQL diagnostic queries against the pre-modeled data to quantify structural flaws and post-modeling to confirm that the Star Schema successfully resolved the data inconsistency issue (conflicting demographics).
- **Analytical Validation Testing (Functional):** Validated that all core business logic implemented via DAX measures (e.g., Repeat Purchase Rate, AOV) returned the numerically correct value. This confirmed that the analytical features behaved as intended.
- **Visualization (UI/UX) Testing:** The final Power BI dashboards were tested to check for proper display, accurate labeling, and smooth, consistent interaction when using slicers and cross-filtering across all four report pages.
- **Security Testing:** Focused on validating the model's structural security, including the functionality of the Row-Level Security (RLS) design (if implemented) and confirming that no sensitive raw data leaked into the presentation layer.

Testing Approach

- **Manual Testing:** Performed regularly by the analyst during development to spot-check calculated fields and visually confirm that charts were displaying logical business trends.

- **Diagnostic SQL Tests:** Used as the primary tool to verify data accuracy and structural integrity *before* and *after* the Power BI modeling phase.
- **Cross-Validation:** Final metrics (KPIs) calculated using DAX were mathematically cross-checked against simple aggregate queries run directly against the source data to guarantee numerical fidelity.
- **End-to-End Testing:** Simulated real user actions by filtering and drilling down through all four dashboards to ensure the strategic insights (RCA findings) remained consistent and supported by the data model.

7.2 TESTING STRATEGY

The testing strategy was focused on ensuring that the core analytical structure and all generated business insights were accurate and reliable. This involved multiple levels of validation across the entire data lifecycle.

- **Unit Testing (Data Cleaning):**
 - Individual transformation steps within the Python cleaning scripts were tested in isolation to verify that they performed their function correctly. This ensures foundational data integrity before modeling.
- **Data Integrity Testing (Modeling Focus):**
 - This was the core strategic test. It verified that the Star Schema model successfully resolved the structural data flaws (conflicting customer demographics). Tests focused on ensuring relationships between the Fact and Dimension Tables were accurate, preventing miscalculation in the final reports.
- **Analytical Validation (Functional Testing):**
 - Complex **DAX measures** (like the Repeat Purchase Rate) were cross-checked against simple aggregate queries run directly in SQL. This rigorous comparison confirmed that the calculated business metrics met the necessary accuracy requirements.
- **Security and Access Testing (RLS Validation):**

- Validation focused on the structural security of the data model. This included ensuring the design supports future Row-Level Security (RLS) implementation and confirming that sensitive raw data attributes were properly anonymized or excluded from the visualization layer.
- **End-to-End Testing (Dashboard Usability):**
 - Manual, scenario-based tests were performed to confirm that the four dashboards worked together seamlessly. Scenarios included cross-filtering between the Product Performance and Customer Insights pages to validate the analytical workflow.
- **Regression Testing:**
 - After any change to the Power Query transformations or core DAX logic, existing key metrics (e.g., Total Revenue, February Sales Dip value) were quickly re-validated to confirm that the functionality and core insights remained intact and unchanged.

This comprehensive strategy ensured that the final Business Intelligence system was structurally sound and provided reliable, high-quality strategic insights.

7.3 TESTING SUITES DESIGN

Test suites in this project were organized collections of test cases designed to thoroughly check the integrity and functional reliability of the analytical model and its outputs. Each suite targets a specific phase of the data pipeline or a core business function to ensure all final metrics are trustworthy.

- **Data Integrity Suite:** This is the most critical suite, designed to validate the structural correction of the raw data. The test cases involve running SQL diagnostic queries to confirm the data inconsistency issue (conflicting demographics) was fully resolved by the Star Schema.
- **Metrics Validation Suite (DAX Accuracy):** This suite targets the core business logic. Test cases involve cross-checking the values of complex DAX measures (like Repeat Purchase Rate and AOV) against simple aggregate calculations run directly in the SQL environment to guarantee numerical fidelity.

- **Root Cause Analysis (RCA) Simulation Suite:** This suite tests the functional and analytical capability of the dashboards. Test cases involve running specific scenarios, such as filtering for the February sales dip and verifying the output confirms the traffic issue, and testing filters on 'Bad' feedback to ensure it isolates the 'Delivered' Order Status.
- **Visualization and Usability Suite:** This suite focuses on the presentation layer (the Power BI Dashboards). Test cases include checking for accurate chart labeling, consistent visual presentation, and ensuring that slicers and filters function seamlessly across all four interconnected reports.

This systematic approach to testing focuses on data trustworthiness and analytical robustness, ensuring the final BI system provides high-quality strategic insights to management.

CHAPTER 8: CONCLUSION AND DISCUSSION

8.1 SELF ANALYSIS OF PROJECT VIABILITIES

This project aims to develop a robust Business Intelligence (BI) system that transforms raw data into strategic insights for the retail sector. After thorough analytical development and validation, the solution demonstrates strong viability on several key fronts.

- **Technical Viability:** The choice of modern, widely-supported analytical technologies like Python (Pandas), SQL, and Microsoft Power BI ensures the system is structurally sound and maintainable. The successful implementation of the Star Schema to resolve data inconsistency proves the core technical fix is scalable and reliable for future data volumes.
- **Operational Viability:** The BI solution automates the analytical process, eliminating reliance on flawed, manual reporting and spreadsheet analysis. It provides management with validated, dynamic, and real-time insight into customer loyalty and operational risks, enabling faster and better strategic decisions.
- **Economic Viability:** By accurately identifying the loyal customer profile and the root cause of the February sales dip, the system enables highly targeted marketing and retention efforts, significantly reducing wasteful spending. The project utilized zero-cost open-source tools, providing a high return on investment with no licensing fees.
- **Schedule Viability:** The project was planned and executed successfully within the proposed 14-week timeline. Effective use of the Iterative Analytical Approach (CRISP-DM) and modular design expedited cleaning, modeling, and final delivery.
- **Analytical Viability:** The features directly address critical analytical blind spots, such as identifying the true source of the 14% 'Bad' feedback (post-delivery quality) and quantifying the Volume vs. Value dynamics of the product portfolio. This provides a strong foundation for future growth and deeper analytical exploration.

In summary, the project is viable analytically, operationally, and economically, and it provides a solid, trustworthy foundation for strategic decision-making in the retail environment.

8.2 PROBLEM ENCOUNTERED AND POSSIBLE SOLUTIONS

The implementation of the analytical solution encountered several structural and data-related challenges that required specific solutions to guarantee the final report's reliability.

- **Data Inconsistency within Single Entities:** The most significant structural problem was the conflicting customer demographic data (Age, Gender, Income) associated with single Customer_IDs in the raw file. This threatened to invalidate all segmentation. The Solution was the implementation of the Star Schema architecture, where the conflicting data was normalized within the dedicated Customer Dimension Table using Power Query logic to ensure a single source of truth for every customer.
- **Handling Various Date and Time Formats:** The raw data contained various date and time formats, making direct, clean time-series analysis impossible. The Solution was addressed early in the Python cleaning phase (Phase 1) by using the `pd.to_datetime()` function with error handling to standardize all dates and subsequently engineer the consistent Time_Slot feature for analysis.
- **Validating Complex Analytical Logic (DAX):** Writing complex DAX measures (like the Repeat Purchase Rate) to ensure accuracy was challenging, given the need for conditional logic and aggregate filtering. The Solution implemented a rigorous Validation Strategy where final DAX values were mathematically cross-checked against simple SQL aggregate queries run on the data model to confirm numerical fidelity.
- **Testing and Scenario Validation:** Testing the analytical validity of the dashboard's Root Cause Analysis (RCA) capability required specialized scenario testing rather than standard software unit tests. The Solution was the development of RCA Simulation Test Suites (manual tests) that confirmed applying specific filters (e.g., Month = February) yielded the expected analytical result (Traffic down, AOV stable), verifying the integrity of the strategic insights.

8.3 SUMMARY OF PROJECT WORK

The Customer Preference and Trend Analytics for Retail Sector project was successfully completed, moving the raw, flawed transactional data through a rigorous analytical pipeline to a final, actionable Business Intelligence (BI) solution.

The entire workflow was structured into four distinct phases. The project successfully completed Phase 1 (Data Cleaning) using Python, and critically solved the data inconsistency issue in Phase 3 (Modeling) by implementing the industry-standard Star Schema in Power BI. This structural fix provides the foundation for accurate analysis, resolving the inherent flaws in the original dataset.

The system's core analytical capabilities include four interconnected dashboards that serve as specialized modules for strategic insight: Customer Insights, Product Performance, Global Sales Overview, and Operational & Feedback. These modules are powered by DAX metrics and facilitate complex Root Cause Analysis (RCA).

Key project successes include quantifying the 68% customer loyalty rate, identifying the cause of the February sales dip as a customer traffic problem, and pinpointing the 14% 'Bad' feedback to post-delivery quality issues. Comprehensive SQL validation and DAX testing were performed to ensure the final solution's reliability and performance. Overall, the project successfully met its intended objectives and provides a scalable, trustworthy analytical foundation for future retail strategy and management.

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CHAPTER 9: LIMITATION AND FUTURE ENHANCEMENT

Limitations:

The project was executed as an analytical development solution with specific constraints. These limitations define the scope of the current system and point toward necessary future work.

- **Static Data Analysis:** The current system operates solely on a historical, static snapshot of the transactional data (df_time_cleaned). It lacks the capability for real-time data ingestion and reporting, which limits its ability to track emerging trends immediately.
- **Descriptive Focus:** The analysis is purely descriptive (What happened?), providing Root Cause Analysis (RCA) on past events. It does not include predictive capabilities (What will happen?), such as forecasting sales or customer churn probability.
- **Analytical Tool Chain:** The solution is dependent on the local installation and configuration of Python and Power BI Desktop, which limits enterprise-wide deployment and collaboration without publishing to a cloud service.
- **Data Structure Scope:** The analytical output is limited to the data provided. The current model does not integrate external factors (e.g., marketing spend, competitor data, or macroeconomic indices) that could enrich the RCA findings.
- **Integration Readiness:** No direct connectors or APIs were developed to integrate the validated Star Schema with external business systems like ERP or accounting software, requiring future development for enterprise use.

Future Enhancements:

Future development should focus on enhancing the project's analytical depth and integrating the solution into a more scalable, predictive environment.

- **Implement Predictive Modeling (ML):** The highest value enhancement would be to implement a Customer Loyalty Prediction Model (e.g., using Logistic

Regression) to forecast which new customers are most likely to become repeat buyers. This moves the project into proactive strategic targeting.

- **Enhance Scalability and Collaboration:** The final Power BI model should be published to the Power BI Service or an equivalent cloud platform. This allows for scalable access, version control of reports, and improved collaboration among all management users.
- **Advanced Analytical Features:** Add advanced features to the dashboards, such as dynamic What-If Analysis (allowing managers to simulate the impact of marketing spend on sales trends) using DAX parameters.
- **Data Pipeline Automation:** Automate the entire analytical workflow, converting the manual Python cleaning scripts into an automated, scheduled process (ETL pipeline) using tools like Azure Data Factory or Python scheduling services.
- **Integrate External Data Sources:** Develop connectors to integrate supplementary data streams, such as marketing campaign spending, to enrich the RCA findings and provide a more complete view of business drivers.

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